TOBYQA: A TIME-AUGMENTED MODEL-BASED METHOD FOR DERIVATIVE-FREE OPTIMIZATION UNDER NOISE AND TEMPORAL DRIFT

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Abstract. Derivative-free optimization (DFO) is challenging when the observation channel varies over time and evaluations are noisy. Conventional model-based methods assume stationary observations; under temporal drift, historical data biases gradient estimates and acceptance tests confound latent-objective decrease with temporal variation. We propose TOBYQA (Time-augmented Optimization BY Quadratic Approximation), a regularized model-based DFO framework that jointly incorporates spatial geometry and temporal variations within a single saddle-point interpolation system. TOBYQA augments the classical least-Frobenius-norm quadratic interpolation system with a linear-in-time drift term and a ridge regularization on the residual kernel, which accommodates noise and ensures well-posedness when the constraint block has full column rank, relaxing the geometric poisedness requirements of classical interpolation. We prove that under affine temporal drift, the recovered gradient is algebraically invariant to the drift rate for any noise scale and sample radius. This property leads to a drift-compensated acceptance test that subtracts the estimated temporal component from the observed reduction. Driven by an adaptive cubic regularization scheme with a closed-form step and a statistical stationarity stopping rule based on the covariance of the gradient estimator, TOBYQA achieves an evaluation complexity of $O(\varepsilon^{-2})$. Numerical experiments on the Moré–Wild benchmark problems demonstrate that TOBYQA achieves substantially higher solve rates than existing derivative-free methods under temporal drift while retaining comparable performance in stationary noisy environments.

Key words. derivative-free optimization, time-varying optimization, noisy optimization, cubic regularization, interpolation models, drift compensation

AMS subject classifications. 90C56, 90C30, 65K05

1. Introduction.

In some black-box optimization problems, objective evaluations are expensive and depend on when they are performed. Examples include aerodynamic wind-tunnel testing and thermal process control, where the underlying physical configuration $\boldsymbol{x} \in \mathbb{R}^n$ may have a fixed optimum while sensor drift or temperature fluctuations change the measurement baseline during operation. Because physical trials consume time and resources, evaluation budgets are limited, and earlier evaluations cannot be repeated under identical conditions. A related setting arises in data-driven materials discovery: in the accelerated design of magnetic high-entropy alloys, Li et al. [12] train machine-learning surrogates of alloy properties as functions of the composition vector $\boldsymbol{x} \in \mathbb{R}^n$ and use the surrogate scores to search the composition space. When such workflows update the surrogate sequentially with newly synthesized batches, the scores supplied to the optimization loop can vary across sequential evaluation rounds.

Motivated by such scenarios, we consider the unconstrained minimization problem

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x}), \tag{1.1}$$

where the latent objective function $f : \mathbb{R}^n \to \mathbb{R}$ is continuous, smooth, and bounded from below by $f_{\text{low}} > -\infty$, but cannot be directly evaluated. Instead, evaluations are obtained sequentially through a drifting, noisy observation channel. Queries are

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indexed by a discrete time $t \in \mathbb{N}$ that increments by one with each oracle evaluation. A query issued at point $\boldsymbol{x}$ and timestamp t returns

$$F(\boldsymbol{x}, t) \;=\; \phi(f(\boldsymbol{x}), t) + \varepsilon(\boldsymbol{x}, t), \tag{1.2}$$

where $\phi : \mathbb{R} \times \mathbb{N} \to \mathbb{R}$ represents an unknown temporal transformation and $\varepsilon(\boldsymbol{x}, t)$ is an independent zero-mean observation noise with variance bounded by σ_ε^2. When $\phi(\cdot, t)$ is strictly increasing with respect to its first argument, the spatial minimizers of f in (1.1) remain invariant over time. However, because evaluations are performed sequentially, observations $F(\boldsymbol{x}_i, t_i)$ collected at distinct time steps reflect different temporal baselines. Finite differences and static interpolation models fitted to such data confound the spatial gradient with temporal variation.

For stationary problems where F depends solely on $\boldsymbol{x}$, derivative-free optimization (DFO) is well established [8, 1, 11]. Model-based methods such as NEWUOA [19] and BOBYQA [20] construct quadratic surrogates over sparse interpolation sets, commonly with $m = 2n + 1$ points, by minimizing the Frobenius norm of the model Hessian change [18]. In noisy settings, Py-BOBYQA [4] provides flexible, noise-aware model management and restart strategies, while the STORM framework [7, 2] establishes global convergence under probabilistically fully-linear models. In dynamic optimization, existing methods mainly target moving optimal trajectories through gradient-based prediction–correction schemes that require temporal derivatives [9, 22], heuristic adaptation in evolutionary algorithms [23], or temporal covariance discounting in Bayesian optimization [14]. These approaches either require explicit derivative information or demand sample budgets that are prohibitive in expensive black-box settings.

Applying classical model-based DFO to time-varying channels raises two difficulties. First, because the interpolation set is accumulated over consecutive iterations, older samples reflect earlier observation baselines; fitting a stationary polynomial surrogate to these space–time observations biases the recovered gradient along the sampling trajectory. Second, the observed difference $F(\boldsymbol{x}_k, t_k) - F(\boldsymbol{x}_k + \boldsymbol{s}_k, t_k + \Delta t)$ combines latent-objective change with the channel variation accumulated over Δt, so a step that improves the latent objective may be rejected while one that worsens it may be accepted. Simple remedies do not resolve this coupling. Periodically re-evaluating the current center $\boldsymbol{x}_k$ to estimate the baseline shift consumes extra function evaluations, advances the clock t—thereby accumulating further drift—and still suffers from a two-point noise variance of $2\sigma_\varepsilon^2$. Likewise, merely appending a temporal term to an exact interpolation model is fragile under noisy sequential sampling: exact interpolation forces the model to fit observation noise $\varepsilon(\boldsymbol{x}_i, t_i)$, while maintaining geometric poisedness of an unregularized space–time KKT system requires dedicated geometry-improving evaluations that again expend budget and advance the temporal drift.

To overcome these difficulties without extra calibration evaluations, we propose TOBYQA (Time-augmented Optimization BY Quadratic Approximation), a regularized model-based DFO algorithm that jointly estimates spatial geometry and temporal drift from $m = 2n + 1$ active samples. Rather than enforcing exact interpolation, TOBYQA formulates a soft-constrained variational problem that augments the constraint block with a temporal column $\boldsymbol{\theta}$ and adds a ridge regularization term $\rho^{-1} I_m$ to Powell's fourth-order kernel A, yielding the strictly positive definite kernel $K = A + \rho^{-1} I_m$. This ridge regularization serves three purposes simultaneously. First, under a rotation-invariant Gaussian prior on the objective Hessian, A is proportional to the spatial covariance of the omitted quadratic Taylor residuals, so solving

the regularized KKT system with $K = A + \rho^{-1} I_m$ performs generalized least-squares estimation that filters observation noise while accounting for unmodeled spatial curvature. Second, because K is strictly positive definite, the saddle-point KKT system is uniquely solvable whenever the space–time constraint matrix X_{aug} has full column rank, relaxing classical geometric poisedness requirements and avoiding extra geometry-maintenance evaluations. Third, because the ridge penalty acts on the dual curvature kernel K rather than shrinking the linear parameters $(c, \boldsymbol{g}, \gamma)$, the primal recovery operator $\Pi = (X_{\mathrm{aug}}^\top K^{-1} X_{\mathrm{aug}})^{-1} X_{\mathrm{aug}}^\top K^{-1}$ remains an exact left inverse of X_{aug} for every $\rho > 0$. Consequently, the recovered spatial gradient is algebraically invariant to affine temporal drift for any noise level and regularization weight, while under general C^2 drift its bias is bounded by $O(L_\mu \tau_{\max}^2)$. Pairing this estimator with a drift-compensated acceptance test, first-in-first-out sample replacement, an $O(n)$ zero-Hessian adaptive cubic regularization step, and a covariance-based stopping rule achieves an $O(\varepsilon^{-2})$ worst-case evaluation complexity above the noise floor and substantially higher solve rates than existing DFO solvers across drifting benchmark channels.

2. Time-Augmented Interpolation Model.

2.1. Soft-Constrained Quadratic Model.

We build on the underdetermined interpolation framework introduced by Powell in NEWUOA [19]. At each iteration, NEWUOA maintains m interpolation points around the current iterate and constructs a quadratic surrogate $Q \approx f$. Because a full quadratic model in $\mathbb{R}^n$ has $(n+1)(n+2)/2$ coefficients, determining all of them by exact interpolation requires $O(n^2)$ evaluations. Powell instead proposed using $m = 2n + 1$ points [18]: the interpolation conditions $Q(\boldsymbol{x}_i) = F_i$ determine the constant and gradient while providing partial curvature information, and the remaining freedom in the Hessian is resolved by solving

$$\min_{Q} \ \|\nabla^2 Q - \nabla^2 Q_{\mathrm{prev}}\|_F \quad \text{subject to} \quad Q(\boldsymbol{x}_i) = F_i, \quad i = 1, \dots, m.$$

This construction presupposes a stationary, noiseless objective. Under the dynamic observation channel (1.2), sequential sampling interleaves spatial displacement with temporal evolution, so a static polynomial model projects temporal shifts onto spatial directions and biases the gradient. Moreover, enforcing $Q(\boldsymbol{x}_i) = F_i$ exactly propagates observation noise directly into the fitted coefficients.

TOBYQA addresses both issues by augmenting the quadratic model with a linear temporal term and replacing exact interpolation with a regularized least-squares fit. At iteration k, let $\boldsymbol{x}_k$ be the current center and t_k its evaluation timestamp. Given the active set $\mathcal{Y}_k = \{(\boldsymbol{x}_i, t_i, F_i)\}_{i=1}^m$ of $m = 2n + 1$ samples, we center the spatial displacements $\boldsymbol{d}_i := \boldsymbol{x}_i - \boldsymbol{x}_k$ and temporal offsets $\theta_i := t_i - t_k$ at $(\boldsymbol{x}_k, t_k)$, and consider the time-augmented model

$$Q_k(\boldsymbol{x}, t) \ = \ c + \boldsymbol{g}^\top (\boldsymbol{x} - \boldsymbol{x}_k) + \tfrac{1}{2} (\boldsymbol{x} - \boldsymbol{x}_k)^\top G \, (\boldsymbol{x} - \boldsymbol{x}_k) + \gamma \, (t - t_k), \tag{2.1}$$

where $c = Q_k(\boldsymbol{x}_k, t_k) \in \mathbb{R}$, $\boldsymbol{g} = \nabla_{\boldsymbol{x}} Q_k(\boldsymbol{x}_k, t_k) \in \mathbb{R}^n$, and $\gamma = \partial_t Q_k(\boldsymbol{x}_k, t_k) \in \mathbb{R}$ represent the local function value, spatial gradient, and instantaneous drift rate at $(\boldsymbol{x}_k, t_k)$, and $G = G^\top \in \mathbb{R}^{n \times n}$ is the symmetric model Hessian. To accommodate observation noise, we fit these parameters by solving

$$\min_{c, \, \boldsymbol{g}, \, G = G^\top, \, \gamma} \ \tfrac{1}{4} \|G\|_F^2 \ + \ \tfrac{\rho}{2} \sum_{i=1}^{m} \big(Q_k(\boldsymbol{x}_i, t_i) - F_i \big)^2, \tag{2.2}$$

where $\rho > 0$ balances curvature regularization against data fitting. Unlike NEWUOA, which penalizes the incremental change $\|\nabla^2 Q - \nabla^2 Q_{\text{prev}}\|_F$ to accumulate second-order information across iterations, problem (2.2) penalizes $\frac{1}{4}\|G\|_F^2$ directly. In a time-varying channel, the previous Hessian $G_{\text{prev}} = \nabla^2 Q_{k-1}$ was fitted around an earlier timestamp t_{k-1} and absorbs unmodeled nonlinear drift from past iterations; penalizing $\|G - G_{\text{prev}}\|_F^2$ would lock these historical drift errors into the KKT right-hand side even after the old samples themselves are discarded from $\mathcal{Y}_k$. Penalizing $\frac{1}{4}\|G\|_F^2$ resets the curvature prior at each iteration and prevents cross-iteration drift accumulation (see the ablation study in Section 5.2, Table 5.5).

Solving (2.2) directly in the primal variables would require optimizing over the $n(n+1)/2$ independent entries of G. To reduce the dimension of the linear system from $O(n^2)$ to $O(n)$, we pass to a dual representation of G by introducing the fitting residuals $\xi_i := F_i - Q_k(\boldsymbol{x}_i, t_i)$ and rewriting (2.2) as the constrained problem

$$\min_{c,\,\boldsymbol{g},\,G=G^\top,\,\gamma,\,\boldsymbol{\xi}} \ \tfrac{1}{4}\|G\|_F^2 + \tfrac{\rho}{2}\sum_{i=1}^{m}\xi_i^2 \quad \text{s.t.} \quad Q_k(\boldsymbol{x}_i, t_i) + \xi_i = F_i, \ \ i = 1, \dots, m, \tag{2.3}$$

with Lagrangian

$$\mathcal{L}(c, \boldsymbol{g}, G, \gamma, \boldsymbol{\xi}, \boldsymbol{\lambda}) \ = \ \tfrac{1}{4}\|G\|_F^2 + \tfrac{\rho}{2}\sum_{i=1}^{m}\xi_i^2 - \sum_{i=1}^{m}\lambda_i\Big(Q_k(\boldsymbol{x}_i, t_i) + \xi_i - F_i\Big). \tag{2.4}$$

Setting the partial derivatives of $\mathcal{L}$ with respect to the primal variables to zero gives the first-order optimality conditions:

$$\nabla_G \mathcal{L} = 0 \implies G \ = \ \sum_{i=1}^{m}\lambda_i \, \boldsymbol{d}_i \, \boldsymbol{d}_i^\top, \tag{2.5}$$

$$\nabla_c \mathcal{L} = 0 \implies \sum_{i=1}^{m}\lambda_i \ = \ 0, \tag{2.6}$$

$$\nabla_{\boldsymbol{g}} \mathcal{L} = 0 \implies \sum_{i=1}^{m}\lambda_i \boldsymbol{d}_i \ = \ \mathbf{0}, \tag{2.7}$$

$$\nabla_\gamma \mathcal{L} = 0 \implies \sum_{i=1}^{m}\lambda_i \theta_i \ = \ 0, \tag{2.8}$$

$$\nabla_{\xi_i} \mathcal{L} = 0 \implies \xi_i \ = \ \tfrac{1}{\rho}\lambda_i. \tag{2.9}$$

Identity (2.5) expresses the $n \times n$ matrix G in terms of the m dual multipliers $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_m)^\top$. Substituting (2.5) and $\xi_i = \lambda_i/\rho$ into $Q_k(\boldsymbol{x}_i, t_i) + \xi_i = F_i$ yields

$$\tfrac{1}{2}\sum_{j=1}^{m}\lambda_j(\boldsymbol{d}_i^\top \boldsymbol{d}_j)^2 + c + \boldsymbol{d}_i^\top \boldsymbol{g} + \gamma\theta_i + \tfrac{1}{\rho}\lambda_i \ = \ F_i, \qquad i = 1, \dots, m. \tag{2.10}$$

Combining (2.10) with the orthogonality conditions (2.6)–(2.8) produces the $(m+n+2) \times (m+n+2)$ saddle-point system

$$\begin{bmatrix} A + \frac{1}{\rho}I_m & \mathbf{1}_m & D & \boldsymbol{\theta} \\ \mathbf{1}_m^\top & 0 & 0 & 0 \\ D^\top & 0 & 0 & 0 \\ \boldsymbol{\theta}^\top & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda} \\ c \\ \boldsymbol{g} \\ \gamma \end{bmatrix} \ = \ \begin{bmatrix} \boldsymbol{F} \\ 0 \\ \mathbf{0} \\ 0 \end{bmatrix}, \tag{2.11}$$

where $A \in \mathbb{R}^{m\times m}$ has entries $A_{ij} = \frac{1}{2}(\boldsymbol{d}_i^\top \boldsymbol{d}_j)^2$, $D \in \mathbb{R}^{m\times n}$ has rows $\boldsymbol{d}_i^\top$, $\boldsymbol{\theta} = (\theta_1, \dots, \theta_m)^\top \in \mathbb{R}^m$, and $\boldsymbol{F} = (F_1, \dots, F_m)^\top$. We denote the augmented constraint block by

$$X_{\text{aug}} := \begin{bmatrix} \mathbf{1}_m & D & \boldsymbol{\theta} \end{bmatrix} \in \mathbb{R}^{m\times(n+2)}. \tag{2.12}$$

2.2. Solvability of the KKT System. To compute $(c, \boldsymbol{g}, \gamma)$ from (2.11) at each iteration, we now establish when the linear system is uniquely solvable and derive an explicit expression for the primal parameters. Writing $K := A + \rho^{-1} I_m$, the coefficient matrix of (2.11) takes the block form

$$M = \begin{bmatrix} K & X_{\text{aug}} \\ X_{\text{aug}}^\top & 0 \end{bmatrix}. \tag{2.13}$$

For a saddle-point matrix of the form (2.13) with symmetric positive definite K, block Gaussian elimination shows that M is nonsingular if and only if the Schur complement $S := X_{\text{aug}}^\top K^{-1} X_{\text{aug}}$ is nonsingular, which holds if and only if X_{aug} has full column rank. We therefore impose the following rank condition on the active set.

DEFINITION 2.1. *The active set $\mathcal{Y}_k = \{(\boldsymbol{x}_i, t_i, F_i)\}_{i=1}^m$ satisfies the augmented column-rank condition if*

$$\operatorname{rank}(X_{\text{aug}}) = n + 2, \tag{2.14}$$

that is, the constant vector $\mathbf{1}_m$, the spatial displacement columns of D, and the temporal offset vector $\boldsymbol{\theta}$ are linearly independent.

PROPOSITION 2.2. *Let $K = A + \rho^{-1} I_m \in \mathbb{R}^{m\times m}$, where $A_{ij} = \frac{1}{2}(\boldsymbol{d}_i^\top \boldsymbol{d}_j)^2$ and $\rho > 0$. Then:*

(i) *K is symmetric and strictly positive definite ($K \succ 0$).*

(ii) *If $\mathcal{Y}_k$ satisfies Definition 2.1, the Schur complement*

$$S := X_{\text{aug}}^\top K^{-1} X_{\text{aug}} \in \mathbb{R}^{(n+2)\times(n+2)} \tag{2.15}$$

is symmetric and strictly positive definite ($S \succ 0$).

(iii) *The matrix M in (2.13) is nonsingular, and the unique primal solution is given by*

$$\begin{bmatrix} c \\ \boldsymbol{g} \\ \gamma \end{bmatrix} = S^{-1} X_{\text{aug}}^\top K^{-1} \boldsymbol{F} =: \Pi \boldsymbol{F}, \tag{2.16}$$

where $\Pi := S^{-1} X_{\text{aug}}^\top K^{-1} \in \mathbb{R}^{(n+2)\times m}$ is the linear recovery operator. The dual variable is $\boldsymbol{\lambda} = K^{-1}(\boldsymbol{F} - X_{\text{aug}} \Pi \boldsymbol{F})$.

Proof. For part (i), let $\boldsymbol{v} \in \mathbb{R}^m$ be nonzero. Since each entry of A satisfies

$$A_{ij} = \tfrac{1}{2}(\boldsymbol{d}_i^\top \boldsymbol{d}_j)^2 = \tfrac{1}{2} \operatorname{tr}\big((\boldsymbol{d}_i \boldsymbol{d}_i^\top)(\boldsymbol{d}_j \boldsymbol{d}_j^\top)\big),$$

A is the Gram matrix of the rank-one matrices $\{\frac{1}{\sqrt{2}} \boldsymbol{d}_i \boldsymbol{d}_i^\top\}_{i=1}^m$ under the Frobenius inner product, so $A \succeq 0$. For any $\rho > 0$, $\boldsymbol{v}^\top K \boldsymbol{v} = \boldsymbol{v}^\top A \boldsymbol{v} + \rho^{-1} \|\boldsymbol{v}\|_2^2 \geq \rho^{-1} \|\boldsymbol{v}\|_2^2 > 0$, which proves $K \succ 0$.

For part (ii), let $\boldsymbol{u} \in \mathbb{R}^{n+2}$ be nonzero. By Definition 2.1, X_{aug} has full column rank, so $\boldsymbol{w} := X_{\text{aug}} \boldsymbol{u} \neq \mathbf{0}$. Because $K \succ 0$, its inverse $K^{-1} \succ 0$. Hence $\boldsymbol{u}^\top S \boldsymbol{u} = \boldsymbol{w}^\top K^{-1} \boldsymbol{w} > 0$, which proves that $S \succ 0$.

For part (iii), the saddle-point matrix M has invertible upper-left block K and invertible Schur complement $-S$. Setting $\boldsymbol{p} := [c, \boldsymbol{g}^\top, \gamma]^\top$, standard block elimination yields $\boldsymbol{\lambda} = K^{-1}(\boldsymbol{F} - X_{\text{aug}}\boldsymbol{p})$. Substituting this into $X_{\text{aug}}^\top \boldsymbol{\lambda} = \mathbf{0}$ gives $S\boldsymbol{p} = X_{\text{aug}}^\top K^{-1}\boldsymbol{F}$, which yields (2.16). ∎

Remark 2.3. In unregularized minimum-Frobenius-norm interpolation [8, 19], the upper-left block A is only positive semidefinite, so nonsingularity of the KKT matrix requires A to be positive definite on the null space of $X_{\text{aug}}^\top$ (quadratic poisedness). Here, the ridge term $\rho^{-1} I_m$ ensures $K \succ 0$ on all of $\mathbb{R}^m$, reducing solvability to the first-order rank condition (2.14).

Geometrically, condition (2.14) requires that the spatial displacements span $\mathbb{R}^n$ and that the temporal offset vector $\boldsymbol{\theta}$ not lie in $\text{span}\{\mathbf{1}_m, D\}$, so that elapsed time is not an affine function of spatial position across the sample set. The next result verifies that the initial coordinate stencil used by TOBYQA automatically satisfies this requirement under sequential evaluation.

COROLLARY 2.4. *Let* $\mathcal{Y}_0 = \{(\boldsymbol{x}_i, t_i, F_i)\}_{i=1}^{2n+1}$ *be the initial active set formed by evaluating the coordinate stencil* $\{\boldsymbol{x}_0\} \cup \{\boldsymbol{x}_0 \pm \Delta_0 \boldsymbol{e}_j\}_{j=1}^{n}$ *at strictly increasing timestamps* $t_1 < t_2 < \cdots < t_{2n+1}$, *with* $\boldsymbol{x}_1 = \boldsymbol{x}_0$ *evaluated at* t_1. *Then* $\text{rank}(X_{\text{aug}}) = n + 2$.

Proof. In the displacement matrix $D \in \mathbb{R}^{(2n+1)\times n}$, the rows for $\boldsymbol{x}_0 \pm \Delta_0 \boldsymbol{e}_j$ are $\pm\Delta_0 \boldsymbol{e}_j^\top$, so the columns of D are linearly independent and $\text{rank}(D) = n$. Moreover, the row of D at the center sample is zero while the corresponding entry of $\mathbf{1}_{2n+1}$ is one, so $\mathbf{1}_{2n+1}$ is independent of the columns of D. Hence $\text{rank}([\mathbf{1}_{2n+1} \;\; D]) = n + 1$.

Suppose, for contradiction, that $\boldsymbol{\theta} \in \text{span}\{\mathbf{1}_{2n+1}, D\}$, so that $\theta_i = \alpha + \boldsymbol{\beta}^\top \boldsymbol{d}_i$ for some $\alpha \in \mathbb{R}$ and $\boldsymbol{\beta} \in \mathbb{R}^n$. The center row gives $\alpha = \theta_1 = 0$. Applying this to the j-th probe pair, whose displacements are $\pm\Delta_0 \boldsymbol{e}_j$ and whose offsets are θ_{2j} and θ_{2j+1}, gives

$$\theta_{2j} = \alpha + \Delta_0 \beta_j, \qquad \theta_{2j+1} = \alpha - \Delta_0 \beta_j \qquad \Longrightarrow \qquad \alpha = \tfrac{1}{2}\big(\theta_{2j} + \theta_{2j+1}\big).$$

Because probe timestamps strictly succeed the center timestamp, $\frac{1}{2}(\theta_{2j} + \theta_{2j+1}) > 0$, contradicting $\alpha = 0$. Therefore $\boldsymbol{\theta} \notin \text{span}\{\mathbf{1}_{2n+1}, D\}$, and $\text{rank}(X_{\text{aug}}) = n + 2$. ∎

2.3. Gradient Estimation under Drift. Having obtained the explicit estimator $[c, \boldsymbol{g}^\top, \gamma]^\top = \Pi \boldsymbol{F}$ in (2.16), we now examine how temporal drift in the observations $\boldsymbol{F}$ affects the recovered spatial gradient $\boldsymbol{g}$ and drift rate γ. From (2.15) and (2.16), the recovery operator $\Pi = S^{-1} X_{\text{aug}}^\top K^{-1}$ is a left inverse of the augmented constraint matrix:

$$\Pi X_{\text{aug}} \;=\; S^{-1}\big(X_{\text{aug}}^\top K^{-1} X_{\text{aug}}\big) \;=\; I_{n+2}. \tag{2.17}$$

Because $\mathbf{1}_m$ is the first column of X_{aug} and $\boldsymbol{\theta}$ is the $(n+2)$-nd column, identity (2.17) implies $\Pi \mathbf{1}_m = \boldsymbol{e}_1$ and $\Pi \boldsymbol{\theta} = \boldsymbol{e}_{n+2}$. Any constant or linear-in-time component in $\boldsymbol{F}$ is therefore mapped entirely into the c- and γ-coordinates, with zero contribution to the gradient coordinates $\boldsymbol{g}$.

THEOREM 2.5. *Suppose the observation channel exhibits affine drift:*

$$F(\boldsymbol{x}, t) \;=\; f(\boldsymbol{x}) + \beta\, t + \varepsilon(\boldsymbol{x}, t), \tag{2.18}$$

for some unknown drift rate $\beta \in \mathbb{R}$. *Let* $\boldsymbol{F}^{\text{stat}} := \boldsymbol{F} - \beta \boldsymbol{t} \in \mathbb{R}^m$ *denote the static observations that would be obtained in the absence of drift, and let* $[c^{\text{stat}}, (\boldsymbol{g}^{\text{stat}})^\top, \gamma^{\text{stat}}]^\top :=$

$\Pi \boldsymbol{F}^{\mathrm{stat}}$ *be the model parameters recovered from* $\boldsymbol{F}^{\mathrm{stat}}$*. If* $\operatorname{rank}(X_{\mathrm{aug}}) = n+2$*, then the parameters recovered by TOBYQA from the drifting observations* $\boldsymbol{F}$ *satisfy:*

$$\boldsymbol{g} \;=\; \boldsymbol{g}^{\mathrm{stat}}, \qquad \gamma \;=\; \gamma^{\mathrm{stat}} + \beta, \qquad c \;=\; c^{\mathrm{stat}} + \beta\, t_k. \tag{2.19}$$

Proof. Let $\boldsymbol{t} = (t_1, \dots, t_m)^\top \in \mathbb{R}^m$ be the vector of query timestamps. Because $t_i = t_k + \theta_i$, we can write $\boldsymbol{t}$ in terms of the columns of $X_{\mathrm{aug}} = [\mathbf{1}_m \;\; D \;\; \boldsymbol{\theta}]$:

$$\boldsymbol{t} \;=\; t_k\, \mathbf{1}_m + \boldsymbol{\theta} \;=\; X_{\mathrm{aug}}\,(t_k \boldsymbol{e}_1 + \boldsymbol{e}_{n+2}), \tag{2.20}$$

where $\boldsymbol{e}_1 = (1, 0, \dots, 0)^\top$ and $\boldsymbol{e}_{n+2} = (0, \dots, 0, 1)^\top$ in $\mathbb{R}^{n+2}$. Applying Π to $\boldsymbol{F} = \boldsymbol{F}^{\mathrm{stat}} + \beta \boldsymbol{t}$ and using (2.17) yields

$$\begin{aligned} \begin{bmatrix} c \\ \boldsymbol{g} \\ \gamma \end{bmatrix} \;=\; \Pi \boldsymbol{F} \;&=\; \Pi \boldsymbol{F}^{\mathrm{stat}} + \beta\, \Pi X_{\mathrm{aug}}\,(t_k \boldsymbol{e}_1 + \boldsymbol{e}_{n+2}) \\ &= \begin{bmatrix} c^{\mathrm{stat}} \\ \boldsymbol{g}^{\mathrm{stat}} \\ \gamma^{\mathrm{stat}} \end{bmatrix} + \beta\,(t_k \boldsymbol{e}_1 + \boldsymbol{e}_{n+2}) \;=\; \begin{bmatrix} c^{\mathrm{stat}} + \beta t_k \\ \boldsymbol{g}^{\mathrm{stat}} \\ \gamma^{\mathrm{stat}} + \beta \end{bmatrix}, \end{aligned}$$

which gives (2.19). ∎

Remark 2.6. Theorem 2.5 is an instance of the Frisch–Waugh–Lovell theorem [10, 13] in weighted least squares: including the temporal offset $\boldsymbol{\theta}$ as an explicit regressor projects out linear temporal trends for any drift rate β, any noise realization, and any sample radius.

When the drift $\mu(t)$ is nonlinear, its Taylor expansion around t_k decomposes into an affine component $\mu(t_k) + \mu'(t_k)(t - t_k)$ plus a second-order remainder. By (2.17), the recovery operator Π still annihilates the affine component, so the estimation bias in $\boldsymbol{g}$ and γ is determined solely by the second-order temporal remainder across the active set.

THEOREM 2.7. *Suppose the observation channel has the form* $F(\boldsymbol{x}, t) = f(\boldsymbol{x}) + \mu(t) + \varepsilon(\boldsymbol{x}, t)$*, where* $\mu \in C^2(\mathbb{R})$ *with* $|\mu''(t)| \le L_\mu$ *for all* t*. Let* $\tau_{\max} := \max_i |t_i - t_k|$ *denote the temporal span of the active set* $\mathcal{Y}_k$*. Then:*

$$\left|\gamma - \gamma^{\mathrm{stat}} - \mu'(t_k)\right| \;\le\; C_\gamma\, L_\mu\, \tau_{\max}, \qquad \left\|\boldsymbol{g} - \boldsymbol{g}^{\mathrm{stat}}\right\|_2 \;\le\; C_g\, L_\mu\, \tau_{\max}^2, \tag{2.21}$$

where $C_\gamma, C_g > 0$ *depend on the normalized active set configuration and* ρ*, but are independent of* $\tau_{\max}$ *and* L_μ*.*

Proof. Taylor expanding $\mu(t_i)$ around t_k gives $\mu(t_i) = \mu(t_k) + \mu'(t_k)\theta_i + r_\mu(t_i)$, where $|r_\mu(t_i)| \le \frac{1}{2} L_\mu \theta_i^2 \le \frac{1}{2} L_\mu \tau_{\max}^2$. In vector form,

$$\boldsymbol{\mu} \;=\; X_{\mathrm{aug}}\big[\mu(t_k)\, \boldsymbol{e}_1 + \mu'(t_k)\, \boldsymbol{e}_{n+2}\big] + \boldsymbol{r}_\mu.$$

By (2.17), Π annihilates the first term, so the perturbation from $\boldsymbol{\mu}$ is governed solely by $\boldsymbol{r}_\mu$:

$$\begin{bmatrix} c - c^{\mathrm{stat}} - \mu(t_k) \\ \boldsymbol{g} - \boldsymbol{g}^{\mathrm{stat}} \\ \gamma - \gamma^{\mathrm{stat}} - \mu'(t_k) \end{bmatrix} \;=\; \Pi\, \boldsymbol{r}_\mu.$$

Writing $\tau := \tau_{\max}$ and factoring the temporal column as $\boldsymbol{\theta} = \tau\, \hat{\boldsymbol{\theta}}$ with $\|\hat{\boldsymbol{\theta}}\|_\infty \le 1$, we set $T := [\mathbf{1}_m \;\; D \;\; \hat{\boldsymbol{\theta}}]$ and $J := \operatorname{diag}(I_{n+1}, \tau)$, so that $X_{\mathrm{aug}} = TJ$. Then $S = J\, \hat{S}\, J$ with

$\hat{S} := T^\top K^{-1} T$. Because K depends only on the spatial displacements and ρ, both K and $\hat{S}$ are independent of τ, and

$$\Pi \boldsymbol{r}_\mu \;=\; J^{-1} \hat{S}^{-1} T^\top K^{-1} \boldsymbol{r}_\mu,$$

where $J^{-1} = \mathrm{diag}(I_{n+1}, \tau^{-1})$. The $(c, \boldsymbol{g})$-components carry the factor 1, while the γ-component carries τ^{-1}. Combining this with $\|\boldsymbol{r}_\mu\|_2 \le \frac{\sqrt{m}}{2} L_\mu \tau^2$ yields (2.21) with $C_\gamma = C_g = \frac{\sqrt{m}}{2} \|\hat{S}^{-1} T^\top K^{-1}\|_2$. ∎

Remark 2.8. Theorem 2.7 shows that the temporal span $\tau_{\max}$ directly controls the gradient error from nonlinear drift. The sample replacement rule in Section 3 removes the oldest non-center sample at each iteration to keep $\tau_{\max}$ bounded.

2.4. Residual Covariance Model. In the optimization algorithm of Section 3, TOBYQA uses only the first-order parameters $(c, \boldsymbol{g}, \gamma)$ from (2.16) to generate trial steps, setting the explicit model Hessian in the step subproblem to zero ($H_k \equiv 0$) because estimating $n(n+1)/2$ curvature entries from $m = 2n+1$ noisy, drifting evaluations amplifies errors of order $O(\sigma_\varepsilon/\delta_k^2)$. A natural question is why we retain the fourth-order kernel $A_{ij} = \frac{1}{2}(\boldsymbol{d}_i^\top \boldsymbol{d}_j)^2$ inside $K = A + \rho^{-1} I_m$ when estimating $(c, \boldsymbol{g}, \gamma)$, and how the regularization weight ρ should be calibrated to the observation noise variance σ_ε^2.

To answer both questions, consider a twice continuously differentiable objective f and its Taylor expansion around $\boldsymbol{x}_k$:

$$f(\boldsymbol{x}_k + \boldsymbol{d}_i) \;=\; f(\boldsymbol{x}_k) + \nabla f(\boldsymbol{x}_k)^\top \boldsymbol{d}_i + \tfrac{1}{2} \boldsymbol{d}_i^\top H \boldsymbol{d}_i + r_3(\boldsymbol{d}_i), \tag{2.22}$$

where $H := \nabla^2 f(\boldsymbol{x}_k)$ and $|r_3(\boldsymbol{d}_i)| = O(\|\boldsymbol{d}_i\|_2^3)$. When the observations $F_i = f(\boldsymbol{x}_i) + \mu(t_i) + \varepsilon_i$ are fitted by the linear terms $c + \boldsymbol{g}^\top \boldsymbol{d}_i + \gamma\theta_i$, the unmodeled spatial residual at sample i is dominated by the quadratic form $q_i := \frac{1}{2} \boldsymbol{d}_i^\top H \boldsymbol{d}_i$. Unlike the independent observation noise ε_i, the curvature residuals q_i and q_j at two displacements $\boldsymbol{d}_i$ and $\boldsymbol{d}_j$ are spatially correlated through the shared Hessian H. The following result shows that under an isotropic prior on H, their covariance matrix is proportional to A.

Proposition 2.9. *Let $H \in \mathbb{R}^{n \times n}$ be a random symmetric matrix drawn from a rotation-invariant ensemble with $\mathbb{E}[H] = 0$ and*

$$\mathbb{E}[H_{ab} H_{cd}] \;=\; \sigma_H^2 \left(\delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}\right), \tag{2.23}$$

where $\sigma_H^2 > 0$ represents the curvature variance scale. Then for any displacement vectors $\boldsymbol{d}_i, \boldsymbol{d}_j \in \mathbb{R}^n$, the covariance between $q_i := \frac{1}{2} \boldsymbol{d}_i^\top H \boldsymbol{d}_i$ and $q_j := \frac{1}{2} \boldsymbol{d}_j^\top H \boldsymbol{d}_j$ is

$$\mathrm{Cov}(q_i, q_j) \;=\; \tfrac{\sigma_H^2}{2} (\boldsymbol{d}_i^\top \boldsymbol{d}_j)^2 \;=\; \sigma_H^2 A_{ij}. \tag{2.24}$$

Proof. Writing $q_i = \frac{1}{2} \sum_{a,b=1}^n H_{ab} d_{i,a} d_{i,b}$, we have $\mathbb{E}[q_i] = 0$, and

$$\begin{aligned}
\mathrm{Cov}(q_i, q_j) \;&=\; \mathbb{E}[q_i q_j] \;=\; \tfrac{1}{4} \sum_{a,b,c,d=1}^{n} \mathbb{E}[H_{ab} H_{cd}]\, d_{i,a} d_{i,b} d_{j,c} d_{j,d} \\
&=\; \tfrac{\sigma_H^2}{4} \sum_{a,b,c,d=1}^{n} \left(\delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}\right) d_{i,a} d_{i,b} d_{j,c} d_{j,d} \\
&=\; \tfrac{\sigma_H^2}{4} \Big((\boldsymbol{d}_i^\top \boldsymbol{d}_j)^2 + (\boldsymbol{d}_i^\top \boldsymbol{d}_j)^2 \Big) \;=\; \tfrac{\sigma_H^2}{2} (\boldsymbol{d}_i^\top \boldsymbol{d}_j)^2 \;=\; \sigma_H^2 A_{ij}.
\end{aligned}$$

∎

By Proposition 2.9, the sum of the curvature residual $\boldsymbol{q} = (q_1, \dots, q_m)^\top$ and the independent observation noise $\boldsymbol{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_m)^\top$ has composite covariance

$$\Sigma_{\text{res}} \;=\; \sigma_H^2 A + \sigma_\varepsilon^2 I_m \;=\; \sigma_H^2\Big(A + \frac{\sigma_\varepsilon^2}{\sigma_H^2} I_m\Big) \;=\; \sigma_H^2\big(A + \rho^{-1} I_m\big) \;=\; \sigma_H^2 K,$$

when the regularization weight is set to the curvature-to-noise ratio

$$\rho \;=\; \frac{\sigma_H^2}{\sigma_\varepsilon^2}. \tag{2.25}$$

Under this covariance model, the generalized least-squares estimator of the linear parameters $[c, \boldsymbol{g}^\top, \gamma]^\top$ is

$$\big(X_{\text{aug}}^\top \Sigma_{\text{res}}^{-1} X_{\text{aug}}\big)^{-1} X_{\text{aug}}^\top \Sigma_{\text{res}}^{-1} \boldsymbol{F} \;=\; \big(X_{\text{aug}}^\top K^{-1} X_{\text{aug}}\big)^{-1} X_{\text{aug}}^\top K^{-1} \boldsymbol{F} \;=\; \Pi \boldsymbol{F},$$

which coincides with the primal KKT solution (2.16). Retaining the kernel A in $K = A + \rho^{-1} I_m$ therefore accounts for the spatial covariance of unmodeled quadratic curvature when estimating $(c, \boldsymbol{g}, \gamma)$, while relation (2.25) provides the calibration rule for ρ used in Section 3.

3. The TOBYQA Algorithm. Given an initial point $\boldsymbol{x}_0 \in \mathbb{R}^n$, an initial stencil radius $\Delta_0 > 0$, and the observation noise variance $R_{\text{obs}} := \sigma_\varepsilon^2$, TOBYQA forms the initial active set $\mathcal{Y}_0$ by evaluating the coordinate stencil $\{\boldsymbol{x}_0\} \cup \{\boldsymbol{x}_0 \pm \Delta_0 \boldsymbol{e}_j\}_{j=1}^n$ at sequential timestamps. By Corollary 2.4, $\mathcal{Y}_0$ satisfies the rank condition $\operatorname{rank}(X_{\text{aug}}) = n + 2$. From the initial function spread $s_F := \max_i F_i - \min_i F_i$ across the stencil, the algorithm matches second- and third-order variations over the radius Δ_0 to set the empirical curvature scale $\widehat{\sigma}_H := s_F / \Delta_0^2$ and the initial cubic regularization weight $\sigma_0 := s_F / \Delta_0^3$, and calibrates the KKT ridge parameter $\rho := \widehat{\sigma}_H^2 / R_{\text{obs}}$ via (2.25).

At iteration k, solving the augmented KKT system (2.11) on $\mathcal{Y}_k$ yields the spatial gradient $\boldsymbol{g}_k$ and drift rate γ_k. Following Section 2.4, we set the explicit model Hessian to zero ($H_k \equiv 0$). With $H_k \equiv 0$, a linear trust-region subproblem would always step to the trust-region boundary regardless of $\|\boldsymbol{g}_k\|_2$; we instead compute the trial step $\boldsymbol{s}_k$ by minimizing the cubic regularization subproblem [5]:

$$\min_{\boldsymbol{s} \in \mathbb{R}^n} m_k(\boldsymbol{s}) \;:=\; \boldsymbol{g}_k^\top \boldsymbol{s} + \tfrac{\sigma_k}{3} \|\boldsymbol{s}\|_2^3, \tag{3.1}$$

where $\sigma_k > 0$ governs the step length. The optimality condition $\nabla m_k(\boldsymbol{s}) = \boldsymbol{g}_k + \sigma_k \|\boldsymbol{s}\|_2 \boldsymbol{s} = \boldsymbol{0}$ gives the unique closed-form minimizer

$$\boldsymbol{s}_k \;=\; -\frac{\boldsymbol{g}_k}{\sqrt{\sigma_k \|\boldsymbol{g}_k\|_2}}, \qquad \|\boldsymbol{s}_k\|_2 \;=\; \sqrt{\frac{\|\boldsymbol{g}_k\|_2}{\sigma_k}}, \tag{3.2}$$

in $O(n)$ operations, so the step norm scales automatically with $\sqrt{\|\boldsymbol{g}_k\|_2}$. We measure the expected decrease by the first-order predicted reduction

$$\text{Pred}_k \;:=\; -\boldsymbol{g}_k^\top \boldsymbol{s}_k \;=\; \|\boldsymbol{g}_k\|_2^{3/2} \sigma_k^{-1/2} \;=\; \|\boldsymbol{g}_k\|_2 \|\boldsymbol{s}_k\|_2, \tag{3.3}$$

which equals $\frac{3}{2}\big(m_k(\boldsymbol{0}) - m_k(\boldsymbol{s}_k)\big)$.

Evaluating the candidate point $\boldsymbol{x}_k + \boldsymbol{s}_k$ at timestamp t_{new} yields $F_{\text{new}} := F(\boldsymbol{x}_k + \boldsymbol{s}_k, t_{\text{new}})$. Over the elapsed interval $t_{\text{new}} - t_k > 0$, a drift rate γ_k shifts F_{new} by approximately $\gamma_k(t_{\text{new}} - t_k)$ and thus shifts the raw difference $F_k - F_{\text{new}}$ by $-\gamma_k(t_{\text{new}} -$

t_k), where $F_k := F(\boldsymbol{x}_k, t_k)$ is the observation at the current center. To remove this baseline shift, we define the *drift-compensated actual reduction*:

$$\mathrm{Ared}_k \;:=\; (F_k - F_{\mathrm{new}}) + \gamma_k\,(t_{\mathrm{new}} - t_k). \tag{3.4}$$

PROPOSITION 3.1. *Under the observation channel $F(\boldsymbol{x}, t) = \phi(f(\boldsymbol{x}), t) + \varepsilon(\boldsymbol{x}, t)$, the drift-compensated reduction (3.4) decomposes as*

$$\mathrm{Ared}_k \;=\; \Delta f(\boldsymbol{x}_k, \boldsymbol{s}_k) \;+\; \mathcal{E}_{\mathrm{drift}} \;+\; \mathcal{E}_{\mathrm{noise}}, \tag{3.5}$$

where:

(i) $\Delta f(\boldsymbol{x}_k, \boldsymbol{s}_k) := f(\boldsymbol{x}_k) - f(\boldsymbol{x}_k + \boldsymbol{s}_k)$ *is the latent objective reduction;*
(ii) $\mathcal{E}_{\mathrm{drift}} := \big[\phi(f(\boldsymbol{x}_k), t_k) - \phi(f(\boldsymbol{x}_k + \boldsymbol{s}_k), t_{\mathrm{new}}) - \Delta f(\boldsymbol{x}_k, \boldsymbol{s}_k)\big] + \gamma_k(t_{\mathrm{new}} - t_k)$ *is the temporal compensation residual;*
(iii) $\mathcal{E}_{\mathrm{noise}} := \varepsilon(\boldsymbol{x}_k, t_k) - \varepsilon(\boldsymbol{x}_k + \boldsymbol{s}_k, t_{\mathrm{new}})$ *is the zero-mean noise difference with variance $2\sigma_\varepsilon^2$.*

In particular, under affine drift $\phi(f(\boldsymbol{x}), t) = f(\boldsymbol{x}) + \beta t$, Theorem 2.5 gives

$$\mathcal{E}_{\mathrm{drift}} \;=\; (\gamma_k - \beta)\,(t_{\mathrm{new}} - t_k) \;=\; \gamma_k^{\mathrm{stat}}\,(t_{\mathrm{new}} - t_k), \tag{3.6}$$

so the drift rate β cancels identically from $\mathcal{E}_{\mathrm{drift}}$.

Proof. Substituting $F(\boldsymbol{x}, t)$ into (3.4) and separating the latent objective difference, temporal residual, and noise terms yields (3.5). Under affine drift, $\phi(f(\boldsymbol{x}_k), t_k) - \phi(f(\boldsymbol{x}_k + \boldsymbol{s}_k), t_{\mathrm{new}}) - \Delta f(\boldsymbol{x}_k, \boldsymbol{s}_k) = -\beta(t_{\mathrm{new}} - t_k)$, while Theorem 2.5 gives $\gamma_k = \gamma_k^{\mathrm{stat}} + \beta$, which yields (3.6). ∎

By Proposition 3.1, the stochastic perturbation in Ared_k is $\mathcal{E}_{\mathrm{noise}}$, which has variance $2R_{\mathrm{obs}}$. When the step is small or approaches a stationary point, the latent reduction $\Delta f(\boldsymbol{x}_k, \boldsymbol{s}_k)$ can be smaller than the noise standard deviation $\sqrt{2R_{\mathrm{obs}}}$, so requiring $\mathrm{Ared}_k > 0$ would reject valid steps under negative noise realizations and inflate σ_k prematurely. TOBYQA therefore separates step acceptance from regularization updates: the trial step is accepted when

$$\mathrm{Ared}_k \;\geq\; -\frac{\kappa_{\mathrm{tol}}}{2}\sqrt{2R_{\mathrm{obs}}}, \tag{3.7}$$

where $\kappa_{\mathrm{tol}} > 0$ is a stationarity tolerance. When (3.7) holds, the center moves to $(\boldsymbol{x}_{k+1}, t_{k+1}, F_{k+1}) = (\boldsymbol{x}_k + \boldsymbol{s}_k, t_{\mathrm{new}}, F_{\mathrm{new}})$, and σ_k is updated via the ratio

$$r_k \;:=\; \frac{\mathrm{Ared}_k}{\mathrm{Pred}_k}, \tag{3.8}$$

setting $\sigma_{k+1} = \sigma_k/\gamma_1$ if $r_k > \eta$ and $\sigma_{k+1} = \sigma_k$ otherwise. If (3.7) fails, the step is rejected, the center remains at $(\boldsymbol{x}_{k+1}, t_{k+1}, F_{k+1}) = (\boldsymbol{x}_k, t_k, F_k)$, and $\sigma_{k+1} = \gamma_2\sigma_k$.

Whether the trial step is accepted or rejected, the new sample $(\boldsymbol{x}_k + \boldsymbol{s}_k, t_{\mathrm{new}}, F_{\mathrm{new}})$ enters the active set to form $\mathcal{Y}_{k+1}$. Maintaining the fixed size $m = 2n + 1$ requires removing one existing point. Classical interpolation methods select the outgoing point based on spatial geometry or distance from the center, which allows chronologically old samples to persist across many iterations. Under nonlinear drift, however, Theorem 2.7 shows that the gradient bias scales as $O(L_\mu \tau_{\max}^2)$ with the temporal span $\tau_{\max} = \max_i |t_i - t_k|$. To keep $\tau_{\max}$ bounded, TOBYQA uses a time-priority (FIFO) replacement rule: the center $\boldsymbol{x}_{k+1}$ is retained, and the chronologically oldest of the remaining $2n$ samples is discarded. Because one non-center point is replaced at every iteration, the age of all non-center points remains bounded by $2n$ evaluation steps.

Finally, after solving the KKT system at each iteration and before computing the trial step, the algorithm checks whether the recovered gradient is statistically distinguishable from observation noise. By Proposition 2.2, under independent observation noise of variance R_{obs}, the primal estimator $[c, \boldsymbol{g}_k^\top, \gamma_k]^\top = \Pi \boldsymbol{F}$ with $\Pi = S^{-1} X_{\mathrm{aug}}^\top K^{-1}$ has covariance

(3.9)
$$\mathrm{Cov}\big(\Pi \boldsymbol{F}\big) \;=\; \Pi\,(R_{\mathrm{obs}} I_m)\,\Pi^\top \;=\; R_{\mathrm{obs}}\, S^{-1}\big(X_{\mathrm{aug}}^\top K^{-2} X_{\mathrm{aug}}\big) S^{-1} \;=:\; R_{\mathrm{obs}}\, P_{\mathrm{norm}}.$$

The trace of the gradient subblock, $\mathrm{tr}\big(R_{\mathrm{obs}}[P_{\mathrm{norm}}]_{gg}\big)$, gives the expected squared norm of the noise-induced gradient error. The algorithm terminates when the recovered gradient norm falls below this noise floor:

$$\|\boldsymbol{g}_k\|_2 \;<\; \kappa_{\mathrm{tol}} \sqrt{\mathrm{tr}\big(R_{\mathrm{obs}}[P_{\mathrm{norm}}]_{gg}\big)}. \tag{3.10}$$

To prevent an ill-conditioned sample geometry from inflating $\mathrm{tr}(P_{\mathrm{norm}})$ and triggering premature termination, condition (3.10) is enforced only when $\mathrm{tr}(P_{\mathrm{norm}}) \le \kappa_{\mathrm{geom}}\,\mathrm{tr}(P_{\mathrm{baseline}})$, where P_{baseline} is the value of P_{norm} on the initial stencil $\mathcal{Y}_0$. The complete procedure is summarized in Algorithm 3.1, with default parameters $\Delta_0 = 2$, $\gamma_1 = 2$, $\gamma_2 = 1.2$, $\eta = 0.75$, $\kappa_{\mathrm{tol}} = 3$, and $\kappa_{\mathrm{geom}} = 5$.

4. Complexity Analysis. We now establish the evaluation complexity of Algorithm 3.1 within the framework of fully-linear interpolation models [8, 7, 4]. We impose the following assumptions on the latent objective function and the interpolation model.

ASSUMPTION 4.1. *The objective function $f : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable with L_g-Lipschitz continuous gradient:*

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\|_2 \;\le\; L_g\,\|\boldsymbol{x} - \boldsymbol{y}\|_2, \qquad \forall\, \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \tag{4.1}$$

and is bounded from below by $f_{\mathrm{low}} > -\infty$ on $\mathbb{R}^n$.

ASSUMPTION 4.2. *There exist constants $0 < \delta_{\mathrm{lo}} \le \delta_{\mathrm{hi}} < \infty$ such that, at every iteration prior to termination,*

$$\delta_{\mathrm{lo}} \;\le\; \delta_k \;\le\; \delta_{\mathrm{hi}}, \qquad \delta_k := \max_{i=1,\dots,m} \|\boldsymbol{x}_i - \boldsymbol{x}_k\|_2. \tag{4.2}$$

ASSUMPTION 4.3. *There exists a constant $\kappa_{eg} > 0$ such that, at every iteration $k \ge 0$, the recovered model gradient $\boldsymbol{g}_k$ satisfies*

$$\|\boldsymbol{g}_k - \nabla f(\boldsymbol{x}_k)\|_2 \;\le\; \kappa_{eg}\,\zeta_k, \tag{4.3}$$

where $\zeta_k := \max\{\delta_k, \sigma_\varepsilon/\delta_k\}$ and δ_k is the active sample radius in (4.2).

Assumptions 4.1–4.3 are standard in derivative-free complexity analysis [8, 4]. Here ζ_k combines the spatial interpolation error $O(\delta_k)$ and the noise amplification $O(\sigma_\varepsilon/\delta_k)$; under affine drift the temporal trend is removed exactly by Theorem 2.5, whereas under nonlinear drift with bounded temporal span $\tau_{\max} = O(n)$ the residual drift error from Theorem 2.7 is absorbed into (4.3).

To analyze step acceptance at iteration k, let $\mathbb{E}_k[\cdot]$ and $\mathbb{P}_k(\cdot)$ denote expectation and probability over the observation noise term $\mathcal{E}_{\mathrm{noise}} = \varepsilon(\boldsymbol{x}_k, t_k) - \varepsilon(\boldsymbol{x}_k + \boldsymbol{s}_k, t_{\mathrm{new}})$ conditional on the iterate $\boldsymbol{x}_k$ and trial step $\boldsymbol{s}_k$. By Proposition 3.1 and $\mathbb{E}_k[\mathcal{E}_{\mathrm{noise}}] = 0$,

Algorithm 3.1 The TOBYQA Algorithm

Require: Initial point $\boldsymbol{x}_0 \in \mathbb{R}^n$, initial radius $\Delta_0 > 0$, noise variance R_{obs}, parameters $\gamma_1, \gamma_2 > 1$, $\eta \in (0,1)$, tolerances $\kappa_{\text{tol}}, \kappa_{\text{geom}} > 0$.

Ensure: Approximate stationary point $\boldsymbol{x}^*$.

Evaluate initial stencil $\{\boldsymbol{x}_0\} \cup \{\boldsymbol{x}_0 \pm \Delta_0 \boldsymbol{e}_j\}_{j=1}^n$ at sequential timestamps to form $\mathcal{Y}_0$; record initial center $(\boldsymbol{x}_0, t_0, F_0)$ with $F_0 := F(\boldsymbol{x}_0, t_0)$.
Set spread $s_F \leftarrow \max_i F_i - \min_i F_i$, curvature scale $\widehat{\sigma}_H \leftarrow s_F/\Delta_0^2$, ridge parameter $\rho \leftarrow \widehat{\sigma}_H^2/R_{\text{obs}}$, and initial cubic weight $\sigma_0 \leftarrow s_F/\Delta_0^3$.
Compute baseline covariance metric P_{baseline} from the KKT system on $\mathcal{Y}_0$.
for $k = 0, 1, 2, \ldots$ **do**
  Construct A, $\boldsymbol{\theta}$, and X_{aug}; solve (2.11) for $(\boldsymbol{\lambda}, c, \boldsymbol{g}_k, \gamma_k)$ and compute P_{norm}.
  if $\|\boldsymbol{g}_k\|_2 < \kappa_{\text{tol}} \sqrt{\operatorname{tr}(R_{\text{obs}}[P_{\text{norm}}]_{gg})}$ **and** $\operatorname{tr}(P_{\text{norm}}) \le \kappa_{\text{geom}} \operatorname{tr}(P_{\text{baseline}})$ **then**
    return $\boldsymbol{x}_k$.
  end if
  Compute trial step $\boldsymbol{s}_k \leftarrow -\boldsymbol{g}_k/\sqrt{\sigma_k \|\boldsymbol{g}_k\|_2}$ and predicted reduction $\text{Pred}_k \leftarrow -\boldsymbol{g}_k^\top \boldsymbol{s}_k$.
  Evaluate $F_{\text{new}} \leftarrow F(\boldsymbol{x}_k + \boldsymbol{s}_k, t_{\text{new}})$ at timestamp t_{new}.
  Compute drift-compensated actual reduction $\text{Ared}_k \leftarrow (F_k - F_{\text{new}}) + \gamma_k(t_{\text{new}} - t_k)$ and ratio $r_k \leftarrow \text{Ared}_k/\text{Pred}_k$.
  if $\text{Ared}_k \ge -\frac{\kappa_{\text{tol}}}{2}\sqrt{2R_{\text{obs}}}$ **then**
    Set $(\boldsymbol{x}_{k+1}, t_{k+1}, F_{k+1}) \leftarrow (\boldsymbol{x}_k + \boldsymbol{s}_k, t_{\text{new}}, F_{\text{new}})$.
    if $r_k > \eta$ **then**
      $\sigma_{k+1} \leftarrow \sigma_k/\gamma_1$.
    else
      $\sigma_{k+1} \leftarrow \sigma_k$.
    end if
  else
    Set $(\boldsymbol{x}_{k+1}, t_{k+1}, F_{k+1}) \leftarrow (\boldsymbol{x}_k, t_k, F_k)$.
    $\sigma_{k+1} \leftarrow \gamma_2 \sigma_k$.
  end if
  Remove from $\mathcal{Y}_k$ the oldest sample not located at $\boldsymbol{x}_{k+1}$.
  Insert $(\boldsymbol{x}_k + \boldsymbol{s}_k, t_{\text{new}}, F_{\text{new}})$ to form $\mathcal{Y}_{k+1}$.
end for

the expected drift-compensated reduction differs from the true objective decrease by the acceptance bias

$$b_k \;:=\; \mathbb{E}_k[\text{Ared}_k] - \big(f(\boldsymbol{x}_k) - f(\boldsymbol{x}_k + \boldsymbol{s}_k)\big) \;=\; \mathbb{E}_k[\mathcal{E}_{\text{drift}}]. \tag{4.4}$$

The following assumption bounds this bias and the noise tail in Ared_k.

ASSUMPTION 4.4. *There exists $\varepsilon > 0$ such that, whenever the current center satisfies $\|\nabla f(\boldsymbol{x}_k)\|_2 \ge \varepsilon$, the acceptance bias in (4.4) obeys*

$$|b_k| \;\le\; \tfrac{1}{4}\,\text{Pred}_k, \tag{4.5}$$

and the noise perturbation $\mathcal{E}_{\text{noise}} = \text{Ared}_k - \mathbb{E}_k[\text{Ared}_k]$ is conditionally sub-Gaussian with variance proxy $2\sigma_\varepsilon^2$.

Condition (4.5) requires the temporal compensation error not to dominate the predicted decrease during non-stationary iterations. Under affine drift, relation (3.6)

gives $b_k = \mathbb{E}_k[\gamma_k^{\mathrm{stat}}](t_{\mathrm{new}} - t_k)$, which reflects any deterministic space–time correlation along the trajectory. Under nonlinear C^2 drift, b_k also includes the temporal truncation error bounded by $O(L_\mu \tau_{\max}^2)$ in Theorem 2.7; because the FIFO replacement rule keeps $\tau_{\max} = O(n)$, condition (4.5) holds whenever the drift curvature L_μ and step interval $t_{\mathrm{new}} - t_k$ are moderate relative to ε.

We first give a sufficient lower bound on the cubic regularization weight σ_k for the trial step (3.2) to reduce the latent objective f.

LEMMA 4.5. *Let Assumption 4.1 hold. If the cubic regularization weight satisfies*

$$\sigma_k \;\geq\; \sigma^*(\boldsymbol{g}_k) \;:=\; \frac{4L_g^2}{\|\boldsymbol{g}_k\|_2}, \tag{4.6}$$

then the objective decrease along the trial step $\boldsymbol{s}_k$ satisfies

$$f(\boldsymbol{x}_k) - f(\boldsymbol{x}_k + \boldsymbol{s}_k) \;\geq\; \tfrac{3}{4}\,\mathrm{Pred}_k \;-\; \|\boldsymbol{g}_k - \nabla f(\boldsymbol{x}_k)\|_2\,\|\boldsymbol{s}_k\|_2. \tag{4.7}$$

Proof. By the descent lemma for L_g-smooth functions (Assumption 4.1), we have

$$\begin{aligned} f(\boldsymbol{x}_k + \boldsymbol{s}_k) \;&\leq\; f(\boldsymbol{x}_k) + \nabla f(\boldsymbol{x}_k)^\top \boldsymbol{s}_k + \tfrac{L_g}{2}\|\boldsymbol{s}_k\|_2^2 \\ &=\; f(\boldsymbol{x}_k) + \boldsymbol{g}_k^\top \boldsymbol{s}_k + \tfrac{L_g}{2}\|\boldsymbol{s}_k\|_2^2 + \big(\nabla f(\boldsymbol{x}_k) - \boldsymbol{g}_k\big)^\top \boldsymbol{s}_k. \end{aligned}$$

Using $\boldsymbol{g}_k^\top \boldsymbol{s}_k = -\mathrm{Pred}_k$ from (3.3) and the Cauchy–Schwarz inequality yields

$$f(\boldsymbol{x}_k) - f(\boldsymbol{x}_k + \boldsymbol{s}_k) \;\geq\; \mathrm{Pred}_k \;-\; \tfrac{L_g}{2}\|\boldsymbol{s}_k\|_2^2 \;-\; \|\boldsymbol{g}_k - \nabla f(\boldsymbol{x}_k)\|_2\,\|\boldsymbol{s}_k\|_2. \tag{4.8}$$

By (3.2)–(3.3) and condition (4.6), the ratio of the quadratic curvature term to Pred_k satisfies

$$\frac{\frac{L_g}{2}\|\boldsymbol{s}_k\|_2^2}{\mathrm{Pred}_k} \;=\; \frac{\frac{L_g}{2}\big(\|\boldsymbol{g}_k\|_2/\sigma_k\big)}{\|\boldsymbol{g}_k\|_2^{3/2}\sigma_k^{-1/2}} \;=\; \frac{L_g}{2\sqrt{\sigma_k\|\boldsymbol{g}_k\|_2}} \;\leq\; \tfrac{1}{4}, \tag{4.9}$$

so $\frac{L_g}{2}\|\boldsymbol{s}_k\|_2^2 \leq \frac{1}{4}\,\mathrm{Pred}_k$. Substituting this bound into (4.8) gives (4.7). ∎

Lemma 4.5 shows that when $\sigma_k \geq \sigma^*(\boldsymbol{g}_k)$ and the gradient error term $\|\boldsymbol{g}_k - \nabla f(\boldsymbol{x}_k)\|_2\|\boldsymbol{s}_k\|_2$ is at most $\frac{1}{4}\mathrm{Pred}_k$, the trial step produces a latent objective decrease of at least $\frac{1}{2}\mathrm{Pred}_k$. Combining this deterministic decrease with Assumption 4.4 bounds the conditional probability that the step is rejected.

LEMMA 4.6. *Let Assumption 4.1 hold, let the residual drift bound (4.5) hold at iteration k, and suppose that*

$$\sigma_k \;\geq\; \sigma^*(\boldsymbol{g}_k) \qquad \textit{and} \qquad \|\boldsymbol{g}_k - \nabla f(\boldsymbol{x}_k)\|_2\,\|\boldsymbol{s}_k\|_2 \;\leq\; \tfrac{1}{4}\,\mathrm{Pred}_k. \tag{4.10}$$

Then the conditional rejection probability obeys

$$\mathbb{P}_k\big(\textit{iteration } k \textit{ is rejected}\big) \;\leq\; e^{-\kappa_{\mathrm{tol}}^2/8} \;=:\; p', \tag{4.11}$$

and $p' \leq \frac{1}{2}$ whenever $\kappa_{\mathrm{tol}} \geq \sqrt{8\ln 2}$.

Proof. By Lemma 4.5 and (4.10), $\Delta f(\boldsymbol{x}_k, \boldsymbol{s}_k) \geq \frac{3}{4}\mathrm{Pred}_k - \|\boldsymbol{g}_k - \nabla f(\boldsymbol{x}_k)\|_2\|\boldsymbol{s}_k\|_2 \geq \frac{1}{2}\mathrm{Pred}_k$. Together with (4.4) and (4.5), this yields

$$\mathbb{E}_k[\mathrm{Ared}_k] \;=\; \Delta f(\boldsymbol{x}_k, \boldsymbol{s}_k) + b_k \;\geq\; \tfrac{1}{2}\mathrm{Pred}_k - \tfrac{1}{4}\mathrm{Pred}_k \;=\; \tfrac{1}{4}\mathrm{Pred}_k \;>\; 0.$$

Since $\mathrm{Ared}_k - \mathbb{E}_k[\mathrm{Ared}_k]$ is conditionally sub-Gaussian with variance proxy $2\sigma_\varepsilon^2 = 2R_{\mathrm{obs}}$, the sub-Gaussian tail bound and the acceptance criterion (3.7) give

$$\begin{aligned}\mathbb{P}_k\big(\mathrm{Ared}_k < -\tfrac{\kappa_{\mathrm{tol}}}{2}\sqrt{2R_{\mathrm{obs}}}\big) \;&\le\; \exp\Big(-\frac{\big(\tfrac{\kappa_{\mathrm{tol}}}{2}\sqrt{2R_{\mathrm{obs}}} + \tfrac{1}{4}\mathrm{Pred}_k\big)^2}{4R_{\mathrm{obs}}}\Big)\\ &\le\; \exp\Big(-\frac{\frac{\kappa_{\mathrm{tol}}^2}{4}\,2R_{\mathrm{obs}}}{4R_{\mathrm{obs}}}\Big) \;=\; e^{-\kappa_{\mathrm{tol}}^2/8},\end{aligned}$$

where the second inequality uses $\mathrm{Pred}_k > 0$. Finally, $p' \le \frac{1}{2}$ holds for $\kappa_{\mathrm{tol}} \ge \sqrt{8\ln 2}$.□

When $\|\nabla f(\boldsymbol{x}_k)\|_2 \ge \varepsilon$ and the model gradient satisfies $\|\boldsymbol{g}_k\|_2 \ge \frac{1}{2}\varepsilon$, the threshold $\sigma^*(\boldsymbol{g}_k) = 4L_g^2/\|\boldsymbol{g}_k\|_2$ is bounded above by $8L_g^2/\varepsilon$. In deterministic cubic regularization, any step with $\sigma_k \ge 8L_g^2/\varepsilon$ is accepted, and since a rejected step multiplies σ_k by γ_2, the regularization weight never exceeds $\sigma_{\max} := \gamma_2(8L_g^2/\varepsilon)$. Under stochastic noise and temporal drift, Lemma 4.6 shows that rejections above $\sigma^*(\boldsymbol{g}_k)$ occur with probability at most $e^{-\kappa_{\mathrm{tol}}^2/8}$, but deriving a deterministic upper bound on σ_k from the random process alone would require uniform lower bounds on $\mathbb{P}_k(r_k > \eta)$ across arbitrary drift profiles. We therefore impose the following confinement condition.

Assumption 4.7. *For the target accuracy $\varepsilon > 0$ with $\varepsilon \le 8L_g^2/\sigma_0$ and $\kappa_{\mathrm{tol}} \ge \sqrt{8\ln 2}$, the regularization parameter satisfies*

$$\sigma^*(\boldsymbol{g}_k) \;\le\; \sigma_k \;\le\; \sigma_{\max} \;:=\; \gamma_2\,\frac{8L_g^2}{\varepsilon}, \tag{4.12}$$

throughout all iterations k prior to termination, i.e., while $\min_{j\le k}\|\nabla f(\boldsymbol{x}_j)\|_2 \ge \varepsilon$.

Under Assumption 4.7, the number of rejected iterations is bounded linearly by the number of accepted iterations along any trajectory.

Proposition 4.8. *Let $\mathcal{S}_K := \{0 \le k < K : \text{iteration } k \text{ is accepted}\}$ and $\mathcal{U}_K := \{0 \le k < K : \text{iteration } k \text{ is rejected}\}$ denote the index sets of successful and unsuccessful iterations among the first K iterations, respectively. Under Assumption 4.7,*

$$|\mathcal{U}_K| \;\le\; \frac{\log\gamma_1}{\log\gamma_2}\,|\mathcal{S}_K| \;+\; \frac{1}{\log\gamma_2}\log\frac{\sigma_{\max}}{\sigma_0}. \tag{4.13}$$

Proof. In Algorithm 3.1, σ_k is divided by γ_1 on accepted iterations with $r_k > \eta$, kept unchanged on accepted iterations with $r_k \le \eta$, and multiplied by γ_2 on rejected iterations. Let $N_1 \le |\mathcal{S}_K|$ be the number of accepted iterations in $\{0,\dots,K-1\}$ with $r_k > \eta$. Taking logarithms and telescoping from $k = 0$ to $K-1$ gives

$$\log\sigma_K \;=\; \log\sigma_0 \;-\; N_1\,\log\gamma_1 \;+\; |\mathcal{U}_K|\,\log\gamma_2.$$

Using $N_1 \le |\mathcal{S}_K|$ and $\sigma_K \le \sigma_{\max}$ from Assumption 4.7, we obtain

$$|\mathcal{U}_K|\,\log\gamma_2 \;=\; \log\frac{\sigma_K}{\sigma_0} + N_1\log\gamma_1 \;\le\; \log\frac{\sigma_{\max}}{\sigma_0} + |\mathcal{S}_K|\,\log\gamma_1.$$

Dividing by $\log\gamma_2 > 0$ yields (4.13). □

We now establish the main iteration and evaluation complexity bound. Without repeated sampling at a fixed point, gradient errors below the observation-noise scale cannot be resolved; the target accuracy ε is therefore required to exceed the noise floor $\varepsilon_{\mathrm{noise}}$, which matches the covariance-based termination threshold (3.10).

THEOREM 4.9. *Let Assumptions 4.1–4.7 hold. Let $\varepsilon > 0$ be any target gradient accuracy with $\varepsilon \le 8L_g^2/\sigma_0$ satisfying*

$$\varepsilon \;\ge\; \varepsilon_{\text{noise}} \;:=\; C_0\,\kappa_{eg}\,\max\Big\{\delta_{\text{hi}},\;\frac{\sigma_\varepsilon}{\delta_{\text{lo}}}\Big\}, \tag{4.14}$$

where $C_0 > 0$ is an absolute constant. Then the number of iterations K_ε (and thus of post-initialization function evaluations) required to reach $\|\nabla f(\boldsymbol{x}_k)\|_2 \le \varepsilon$ satisfies

$$K_\varepsilon \;=\; O\!\left(\frac{L_g\,\big(f(\boldsymbol{x}_0) - f_{\text{low}}\big)}{\varepsilon^2}\right). \tag{4.15}$$

Proof. For any accepted step $k \in \mathcal{S}_K$ prior to termination, $\|\nabla f(\boldsymbol{x}_k)\|_2 \ge \varepsilon$. By Assumption 4.3 and condition (4.14),

$$\|\boldsymbol{g}_k - \nabla f(\boldsymbol{x}_k)\|_2 \;\le\; \kappa_{eg}\,\zeta_k \;\le\; \frac{\varepsilon}{C_0} \;\le\; \frac{\|\nabla f(\boldsymbol{x}_k)\|_2}{C_0}.$$

Setting $C_0 \ge 8$ gives $\|\boldsymbol{g}_k - \nabla f(\boldsymbol{x}_k)\|_2 \le \frac{1}{8}\|\nabla f(\boldsymbol{x}_k)\|_2$, and the reverse triangle inequality yields

$$\|\boldsymbol{g}_k\|_2 \;\ge\; \|\nabla f(\boldsymbol{x}_k)\|_2 - \|\boldsymbol{g}_k - \nabla f(\boldsymbol{x}_k)\|_2 \;\ge\; \tfrac{7}{8}\|\nabla f(\boldsymbol{x}_k)\|_2 \;\ge\; \tfrac{1}{2}\,\varepsilon.$$

Consequently, the gradient error term in (4.7) satisfies

$$\begin{aligned}\|\boldsymbol{g}_k - \nabla f(\boldsymbol{x}_k)\|_2\,\|\boldsymbol{s}_k\|_2 \;&\le\; \frac{\varepsilon}{C_0}\,\|\boldsymbol{s}_k\|_2 \;\le\; \frac{2\|\boldsymbol{g}_k\|_2}{C_0}\,\|\boldsymbol{s}_k\|_2 \\ &=\; \frac{2}{C_0}\,\mathrm{Pred}_k \;\le\; \tfrac{1}{4}\,\mathrm{Pred}_k.\end{aligned}$$

Since $\sigma_k \ge \sigma^*(\boldsymbol{g}_k)$ by Assumption 4.7, Lemma 4.5 implies that

$$f(\boldsymbol{x}_k) - f(\boldsymbol{x}_k + \boldsymbol{s}_k) \;\ge\; \tfrac{3}{4}\,\mathrm{Pred}_k - \|\boldsymbol{g}_k - \nabla f(\boldsymbol{x}_k)\|_2\,\|\boldsymbol{s}_k\|_2 \;\ge\; \tfrac{1}{2}\,\mathrm{Pred}_k.$$

On an accepted iteration $k \in \mathcal{S}_K$, we have $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \boldsymbol{s}_k$. Using $\mathrm{Pred}_k = \|\boldsymbol{g}_k\|_2^{3/2}\sigma_k^{-1/2} \ge (\varepsilon/2)^{3/2}\sigma_{\max}^{-1/2}$ together with $\sigma_{\max} = \gamma_2\frac{8L_g^2}{\varepsilon}$ from (4.12) gives

$$f(\boldsymbol{x}_k) - f(\boldsymbol{x}_{k+1}) \;\ge\; \tfrac{1}{2}\left(\frac{\varepsilon}{2}\right)^{3/2}\sqrt{\frac{\varepsilon}{8\gamma_2 L_g^2}} \;=\; c\,\frac{\varepsilon^2}{L_g}, \tag{4.16}$$

where $c := (16\sqrt{\gamma_2})^{-1} > 0$.

On rejected iterations $k \in \mathcal{U}_K$, the center is unchanged ($\boldsymbol{x}_{k+1} = \boldsymbol{x}_k$), so $f(\boldsymbol{x}_k) - f(\boldsymbol{x}_{k+1}) = 0$. Summing (4.16) over all iterations up to K yields

$$f(\boldsymbol{x}_0) - f_{\text{low}} \;\ge\; f(\boldsymbol{x}_0) - f(\boldsymbol{x}_K) \;=\; \sum_{k\in\mathcal{S}_K}\big(f(\boldsymbol{x}_k) - f(\boldsymbol{x}_{k+1})\big) \;\ge\; |\mathcal{S}_K|\,c\,\frac{\varepsilon^2}{L_g},$$

and therefore

$$|\mathcal{S}_K| \;\le\; \frac{1}{c}\,\frac{L_g\,\big(f(\boldsymbol{x}_0) - f_{\text{low}}\big)}{\varepsilon^2} \;=\; O\!\left(\frac{L_g\,\big(f(\boldsymbol{x}_0) - f_{\text{low}}\big)}{\varepsilon^2}\right). \tag{4.17}$$

TABLE 5.1

Observation drift channels used in the benchmark experiments. Here $s_0 = \max(|f(\boldsymbol{x}_0)|, 1)$*,* t *is the sequential evaluation index, and the Lorenz trajectory is generated with standard parameters* $(s, r, b) = (10, 28, 8/3)$*, initial state* $(1, 1, 1)$*, and integration step* 0.01*.*

Channel	Analytical Form $\phi(f(\boldsymbol{x}), t)$	Classification
Static	$f(\boldsymbol{x})$	Baseline ($\beta = 0$)
LinearDrift	$f(\boldsymbol{x}) + 0.01\, s_0\, t$	Affine drift (Theorem 2.5)
PeriodicAdd	$f(\boldsymbol{x}) + 0.05\, s_0 \sin(2\pi t/50)$	C^2 bounded drift (Theorem 2.7)
ChaoticDrift	$f(\boldsymbol{x}) + 0.03\, s_0\, \mathrm{Lorenz}_x(t)$	Chaotic dynamic drift
Hetero	$f(\boldsymbol{x}) + 0.01\, s_0\, t\, (1 + \tanh \|\boldsymbol{x} - \boldsymbol{x}_0\|)$	Spatially coupled drift
MultCoupling	$f(\boldsymbol{x}) \cdot (1 + 10^{-4} t)$	Multiplicative drift
PeriodicMult	$f(\boldsymbol{x}) \cdot (1 + 0.05 \sin(2\pi t/50))$	Multiplicative oscillatory drift

By Proposition 4.8 and $\sigma_{\max}/\sigma_0 = \gamma_2(8L_g^2)/(\sigma_0\varepsilon) = O(\varepsilon^{-1})$, the number of rejected iterations satisfies

$$|\mathcal{U}_K| \;\le\; \frac{\log \gamma_1}{\log \gamma_2}\, |\mathcal{S}_K| \;+\; O(\log(\varepsilon^{-1})).$$

Combining the bounds on $|\mathcal{S}_K|$ and $|\mathcal{U}_K|$ gives

$$K_\varepsilon \;=\; |\mathcal{S}_K| + |\mathcal{U}_K| \;\le\; \left(1 + \frac{\log \gamma_1}{\log \gamma_2}\right) |\mathcal{S}_K| \;+\; O(\log(\varepsilon^{-1})) \;=\; O\left(\frac{L_g\,(f(\boldsymbol{x}_0) - f_{\mathrm{low}})}{\varepsilon^2}\right).$$

Since Algorithm 3.1 performs one function evaluation per iteration after the initial $2n + 1$ stencil evaluations, the total evaluation complexity is $N_{\mathrm{eval}} = 2n + 1 + K_\varepsilon = O(L_g(f(\boldsymbol{x}_0) - f_{\mathrm{low}})\varepsilon^{-2})$. ∎

Remark 4.10. The $O(\varepsilon^{-2})$ rate in Theorem 4.9 matches the optimal complexity of first-order methods on $C^{1,1}$ nonconvex objectives [3]. With a linear sample budget $m = 2n + 1$, the active set provides a fully-linear gradient model rather than an accurate Hessian. Achieving the $O(\varepsilon^{-3/2})$ rate of second-order adaptive cubic regularization [17, 5, 6] requires $\|(H_k - \nabla^2 f(\boldsymbol{x}_k))\boldsymbol{s}_k\|_2 = O(\|\boldsymbol{s}_k\|_2^2)$ [5, AM.4], which in the derivative-free setting demands $O(n^2)$ interpolation points. Under temporal drift, maintaining $O(n^2)$ sequential samples stretches the temporal span to $\tau_{\max} = O(n^2)$ and amplifies the nonlinear drift bias of Theorem 2.7 to $O(L_\mu n^4)$. By operating with $m = 2n + 1$ points and $H_k \equiv 0$, TOBYQA keeps $\tau_{\max} = O(n)$ and achieves the first-order complexity rate.

5. Numerical Experiments. We evaluate the empirical performance of TOBYQA on the Moré–Wild benchmark collection [15] across both stationary and drifting observation channels. The testbed comprises 65 scalable unconstrained problems evaluated at dimensions $n \in \{6, 10, 20\}$ across 5 independent noise realizations, for a total of 975 problem instances per channel. Each problem is evaluated through one of the seven observation channels summarized in Table 5.1. Channel amplitudes and observation noise standard deviations are scaled by the initial function magnitude $s_0 := \max(|f(\boldsymbol{x}_0)|, 1)$, with $\sigma_\varepsilon = 10^{-3} s_0$. The evaluation index advances as $t \leftarrow t + 1$ upon each function evaluation across all solvers.

We compare TOBYQA against representative classical and modern derivative-free solvers: Powell's unconstrained interpolation solver NEWUOA [19] executed via the PDFO library [21], the noise-aware trust-region solver Py-BOBYQA [4], the Nelder–Mead simplex search method [16], and a quasi-Newton BFGS implementation using forward-difference gradient approximations ($h = 10^{-4}$). Every solver

TABLE 5.2
Solve rates (%). Budget: 1500 evaluations per instance.

Solver	$\tau = 10^{-1}$			$\tau = 10^{-3}$			$\tau = 10^{-5}$			$\tau = 10^{-7}$		
	n=6	n=10	n=20	n=6	n=10	n=20	n=6	n=10	n=20	n=6	n=10	n=20
TOBYQA	91.4	88.6	87.3	70.2	59.9	41.7	34.4	32.0	26.2	28.3	27.7	24.0
NEWUOA	71.6	60.8	55.9	20.5	17.5	13.8	8.3	10.4	9.5	4.9	5.8	7.8
Py-BOBYQA	74.8	60.7	41.4	46.9	27.3	10.3	24.4	14.2	7.0	21.1	12.5	6.5
Nelder–Mead	34.2	19.8	11.7	8.1	3.6	3.0	3.2	2.2	2.8	1.4	1.0	1.2
BFGS (FD)	13.7	10.0	0.9	1.0	0.4	0.7	0.5	0.4	0.7	0.4	0.4	0.7

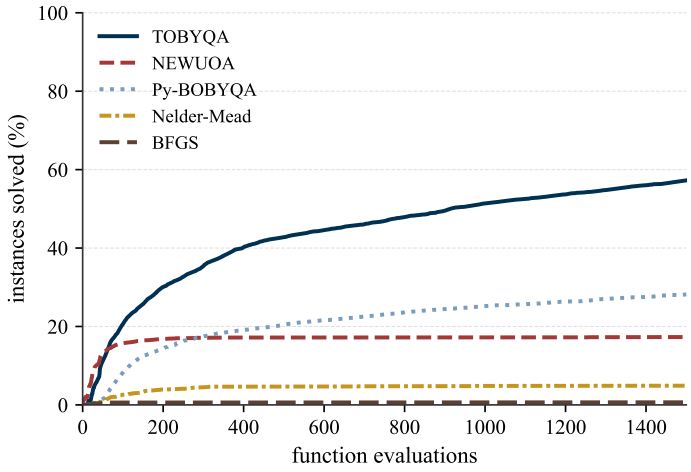


FIG. 5.1. *Data profiles at tolerance $\tau = 10^{-3}$: fraction of benchmark instances solved as a function of the number of function evaluations, pooled across all test problems, dimensions, and drift channels.*

is allocated a maximum budget of 1500 evaluations per instance. Following Moré and Wild [15], performance is measured on the noise-free latent objective f, and an instance is declared solved at accuracy tolerance $\tau \in \{10^{-1}, 10^{-3}, 10^{-5}, 10^{-7}\}$ if $\min_k f(\boldsymbol{x}_k) \le f_L + \tau(f(\boldsymbol{x}_0) - f_L)$, where f_L is the smallest function value found across all solvers.

5.1. Benchmark Comparisons. Table 5.2 reports the solve rates across dimensions and convergence tolerances, while Figures 5.1 and 5.2 display the corresponding data profiles and accuracy decay curves. Across all evaluated dimensions, the performance margin of TOBYQA over the comparison solvers widens as the accuracy tolerance tightens from $\tau = 10^{-1}$ to 10^{-7}. At loose tolerances, coarse initial steps can occasionally reduce the objective before channel drift accumulates; at high precision ($\tau \le 10^{-5}$), however, further progress requires resolving per-step objective reductions smaller than the per-step channel shift $\beta\Delta t$, where uncompensated methods stall. As reflected in the data profiles (Figure 5.1), the baseline solvers exhaust their progress within the first 200–300 evaluations, whereas TOBYQA continues to reduce the objective throughout the 1500-evaluation budget.

The per-channel breakdown in Table 5.3 separates the effects of observation noise and temporal drift. On the Static channel (which still contains observation

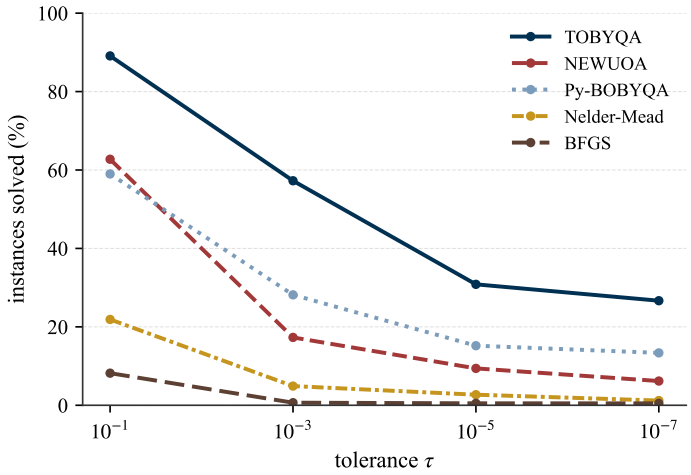


FIG. 5.2. *Solve rates as a function of convergence tolerance* $\tau \in [10^{-7}, 10^{-1}]$*, pooled across all benchmark configurations.*

TABLE 5.3
Solve rates (%) at $\tau = 10^{-3}$ *broken down by drift regime, pooled over all* n *and seeds.*

Solver	Chaotic	Hetero	Linear	MultCoupling	PeriodicAdd	PeriodicMult	Static
TOBYQA	56.0	50.1	64.9	56.5	56.2	59.2	57.9
NEWUOA	10.4	11.6	10.1	29.4	14.7	20.1	24.9
Py-BOBYQA	14.2	3.0	4.0	55.4	19.3	42.1	59.3
Nelder–Mead	1.1	0.0	0.6	12.3	3.1	6.3	10.9
BFGS (FD)	1.9	0.0	0.4	0.3	0.8	0.8	0.3

noise $\sigma_\varepsilon = 10^{-3}s_0$), the unregularized interpolation solver NEWUOA (24.9%) already suffers from noise sensitivity, while TOBYQA (57.9%) matches the noise-aware solver Py-BOBYQA (59.3%), confirming that adding a temporal degree of freedom incurs no practical loss when drift is absent. Once additive temporal drift is introduced, however, noise robustness alone is insufficient: Py-BOBYQA drops from 59.3% on Static to 4.0% on LinearDrift and 3.0% on Hetero, while TOBYQA achieves 64.9% and 50.1%. Because sample points are evaluated sequentially, an unmodeled trend βt tilts the fitted stationary quadratic surrogate along the chronological sampling path; the recovered spatial gradient absorbs this temporal trend and biases trial steps away from $-\nabla f(\boldsymbol{x}_k)$. Finite-difference BFGS fails even more severely: evaluating forward differences over consecutive steps introduces an explicit bias $(F(\boldsymbol{x} + h\boldsymbol{e}_j, t+1) - F(\boldsymbol{x}, t))/h \approx \partial f/\partial x_j + \beta/h$. With stepsize $h = 10^{-4}$ and drift rate $\beta = 10^{-2}s_0$, the drift bias $\beta/h \sim 10^2 s_0$ overwhelms the true gradient and corrupts the quasi-Newton updates.

This surrogate distortion also accounts for the evaluation counts and early termination rates in Table 5.4. In classical trust-region methods such as NEWUOA, the trust-region radius Δ_k contracts whenever the observed reduction falls below the predicted decrease ($r_k < \eta$). Under temporal drift, candidate evaluations are penalized by the accumulated channel shift and are repeatedly rejected. As a result, the trust-region radius shrinks monotonically to its stopping threshold ρ_{end} without moving the center, triggering false convergence on 99.0% of runs at a median of only 185

TABLE 5.4
Evaluation budget utilization and early termination rates across all benchmark instances.

Solver	Median Evals Used	% of Budget	% Early Termination
TOBYQA	1500	100.0	8.5
NEWUOA	185	12.3	99.0
Py-BOBYQA	1500	100.0	24.2
Nelder–Mead	1500	100.0	0.0
BFGS (FD)	180	12.0	99.6

TABLE 5.5
Ablation of TOBYQA components: solve rates (%) at $\tau = 10^{-3}$ by drift regime, pooled over all n and seeds.

Solver	Chaotic	Hetero	Linear	MultCoupling	PeriodicAdd	PeriodicMult	Static
TOBYQA	56.0	50.1	64.9	56.5	56.2	59.2	57.9
+ interpolated H	47.4	55.1	66.4	61.2	50.6	63.9	59.5
– drift comp. (γ_k=0)	9.6	7.1	8.5	55.5	12.7	46.2	56.0
– curvature kernel A	29.0	30.1	39.9	35.5	34.6	36.3	34.5
+ Powell update	11.9	18.7	16.9	14.4	13.4	14.8	14.5
frozen σ	39.1	36.3	38.4	37.6	38.3	36.1	36.2

evaluations. In contrast, TOBYQA prevents premature step contraction by subtracting the estimated temporal trend from the actual reduction (Proposition 3.1), while the covariance-based stopping criterion (3.10) terminates only when the recovered gradient norm falls below the statistical noise floor.

5.2. Ablation Studies. Table 5.5 presents an ablation study isolating the effects of drift compensation, kernel structure, curvature updates, and adaptive regularization across the seven drift channels. Suppressing drift compensation in the acceptance test ($\gamma_k = 0$ in (3.4)) demonstrates that gradient decoupling alone is insufficient: although the spatial gradient $\boldsymbol{g}_k$ remains invariant to affine drift (Theorem 2.5), the acceptance test (3.7) still evaluates raw function differences. A genuine descent step ($\Delta f > 0$) is rejected whenever the channel drift over the evaluation interval exceeds the objective decrease ($\beta \Delta t > \Delta f$), which drives σ_k upward and stalls the iterate. Consequently, while the Static solve rate is nearly unchanged (56.0% vs. 57.9%), solve rates drop to 7.1%–12.7% across the additive drift channels.

Replacing the curvature prior kernel $K = A + \rho^{-1} I_m$ with an isotropic ridge penalty $K = \rho^{-1} I_m$ reduces solve rates from 57.9% to 34.5% on Static and from 64.9% to 39.9% on LinearDrift. Consistent with Proposition 2.9, omitting $A_{ij} = \frac{1}{2}(\boldsymbol{d}_i^\top \boldsymbol{d}_j)^2$ treats unmodeled quadratic residuals as spatially uncorrelated noise rather than structured curvature variation. Incorporating Powell's incremental curvature update $\|\nabla^2 Q - \nabla^2 Q_{\text{prev}}\|_F^2$ degrades solve rates even further to 11.9%–18.7% across all channels: under temporal drift, the Hessian inherited from iteration $k-1$ reflects an outdated temporal baseline, and propagating it across iterations compounds historical drift errors. Finally, fixing the cubic regularization weight to its initial value $\sigma_k \equiv \sigma_0$ ($\gamma_1 = \gamma_2 = 1$) reduces solve rates to 36.1%–39.1%, confirming the necessity of adapting σ_k to local curvature and step acceptance feedback.

Table 5.6 compares the default zero-Hessian Cauchy step ($H_k \equiv 0$) against a trial step computed with the interpolated Hessian $H_k = \sum_{i=1}^{m} \lambda_i \boldsymbol{d}_i \boldsymbol{d}_i^\top$. At $n = 6$, the zero-Hessian default achieves higher solve rates across all tolerances and budgets

TABLE 5.6

Solve rates (%) of the zero-Hessian default versus the interpolated Hessian, by dimension and tolerance. The first three column groups use the main budget of 1500 *evaluations; the last two use scaled budgets (*$n = 6$ *with* 450 *evaluations,* $n = 20$ *with* 5000*). Pooled over all* 65 *problems,* 7 *drift channels, and* 5 *seeds.*

	$n = 6$		$n = 10$		$n = 20$		$n = 6$, $b = 450$		$n = 20$, $b = 5000$	
Solver	10^{-3}	10^{-5}	10^{-3}	10^{-5}	10^{-3}	10^{-5}	10^{-3}	10^{-5}	10^{-3}	10^{-5}
TOBYQA ($H_k \equiv 0$)	70.2	34.4	59.9	32.0	41.7	26.2	83.3	64.3	67.1	45.9
+ interpolated H	62.4	24.0	60.1	29.9	50.7	38.9	67.3	44.9	70.8	59.7

(70.2% vs. 62.4% at $\tau = 10^{-3}$ under budget 1500, and 83.3% vs. 67.3% under budget 450). With $m = 2n+1$ samples, second-order difference quotients amplify observation noise by $O(\sigma_\varepsilon/\delta_k^2)$ over an underdetermined subspace ($n(n+1)/2 = 21$ parameters vs. $n - 1 = 5$ curvature degrees of freedom). Forming trial steps from H_k therefore inverts noise-corrupted directional curvatures, whereas the Cauchy step (3.2) relies only on the fully-linear gradient $\boldsymbol{g}_k$.

At $n = 20$, the pooled solve rate in Table 5.6 favors the interpolated Hessian (50.7% vs. 41.7% at budget 1500, and 70.8% vs. 67.1% at budget 5000). However, the per-channel breakdown in Table 5.5 shows that this gain is restricted to stationary, affine, and multiplicative channels (59.5% vs. 57.9% on Static and 66.4% vs. 64.9% on LinearDrift), whereas on nonlinear additive drift channels the interpolated Hessian drops from 56.0% to 47.4% on ChaoticDrift and from 56.2% to 50.6% on PeriodicAdd (at $n = 20$ and budget 5000, solve rates fall to 54.5% and 49.8% versus 72.1% and 75.7% for $H_k \equiv 0$). Under nonlinear drift, the second-order temporal remainder $\boldsymbol{r}_\mu$ from Theorem 2.7 enters the dual multipliers $\boldsymbol{\lambda} = K^{-1}(\boldsymbol{F} - X_{\text{aug}}\Pi\boldsymbol{F})$ directly, corrupting the eigenvalues of $H_k = \sum_{i=1}^m \lambda_i \boldsymbol{d}_i \boldsymbol{d}_i^\top$ and distorting second-order steps. For general time-varying channels with unknown drift structure, setting $H_k \equiv 0$ avoids curvature corruption from nonlinear drift while maintaining strong performance on stationary problems.

6. Conclusion. Standard model-based derivative-free methods construct local interpolation models from sequential function evaluations under the assumption that the objective is stationary. When observations are corrupted by both random noise and temporal drift, unmodeled drift biases the fitted gradient along the sampling trajectory and causes trust-region methods to reject descent steps and contract their radius prematurely. In this paper, we have proposed TOBYQA to address this setting using $2n + 1$ active sample points. By augmenting a regularized minimum-Frobenius-norm quadratic model with a linear temporal term, TOBYQA decouples the spatial gradient from affine drift and bounds the gradient bias under smooth nonlinear drift in terms of the temporal span of the sample set, which is kept at $O(n)$ evaluations by first-in-first-out sample replacement. Rather than forming an explicit second-order step Hessian from $2n + 1$ noisy, drifting evaluations, the algorithm retains Powell's fourth-order kernel in the KKT system as a spatial covariance weight for unmodeled quadratic residuals, and pairs the resulting generalized least-squares gradient and drift estimators with a zero-Hessian adaptive cubic regularization step, a drift-compensated acceptance test, and a covariance-based stopping rule. Together, these components achieve an $O(\varepsilon^{-2})$ worst-case evaluation complexity above the statistical noise floor.

On the Moré–Wild benchmark collection, TOBYQA performs comparably to PyBOBYQA on stationary noisy problems and achieves substantially higher solve rates

than existing derivative-free solvers when the observation channel drifts over time. Our ablation experiments confirm that drift compensation in the acceptance test, the fourth-order curvature kernel in the KKT system, and adaptive cubic regularization are all required to prevent premature stagnation, whereas propagating incremental Hessian updates across iterations or computing trial steps from an explicit interpolated Hessian at $m = 2n + 1$ exposes the solver to noise amplification and nonlinear drift distortion. Future work includes incorporating higher-order or localized temporal basis functions to track rapidly oscillating channels without degrading the conditioning of the KKT system, extending the method to bound and linear constraints, and generalizing the framework from drifting observation channels to time-varying objectives $f(\boldsymbol{x}, t)$ whose minimizers move over time.

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